

UAT Test Plan for CHD Bayesian Predictor Prototype

Scope

Objectives and business requirements

- **Goal:** To test the usability of a Bayesian network application designed to calculate the probability of Coronary Heart Disease (CHD) based on patient symptoms. The testing is strictly focused on software usability and user experience, not the medical knowledge of the user or the medical accuracy of the model.
- **Target Audience / Use Case:** The application is intended primarily for educational purposes, such as teaching medical students.
- **Success Metrics:** Success is measured by the user's ability to easily navigate the tool, understand the AI explainer, and successfully generate a disease risk output.

Scope

- **What we are testing:** First impressions, data entry usability, output interpretation (AI explainer and percentages), interactive "what-if" scenarios, comprehension of breakdown metrics, and missing data/error handling.
- **What we are not testing:** The medical accuracy of the Bayesian network itself.

Testing team

- **Project Owner / UAT Coordinator:** Handles UAT coordination, guides external testers, manages the testing process, and analyses the final results.
- **External Participants :** Act as User Testers. Responsible for executing test cases, providing "think aloud" feedback on usability, and completing pre and post test questionnaires.

Milestones and deliverables

- **Design & testing process:** Completed.
- **Staging environment:** Completed.
- **Training / Briefing:** Completed via pre-test introductions.
- **UAT Execution:** Completed.
- **Reporting:** Completed via Post-Test Questionnaires and observation logs.

Environmental requirements

Hardware requirements

- **Device:** Testing must be conducted exclusively on a computer (desktop or laptop).
- **Constraints:** The application must not be tested on a mobile phone, as the graphical elements are quite large and are not currently optimized for mobile displays.

Software requirements

- A standard, modern web browser to access the prototype.

Features to be tested

General Pass/Fail Criteria

- **Pass:** A disease risk output percentage is successfully calculated and displayed on the screen based on the user's actions.
- **Fail:** No disease risk output is displayed (e.g., the system crashes, hangs, or fails to render the final percentage).

Feature 1: First Impressions & Interface Clarity

- **Test Case:** Assess the initial interface before any data is entered.
- **Instructions:** Ask the user to observe the page and "think aloud" about their first impressions and whether it is clear what they are supposed to do.

Feature 2: Data Entry (Usability & Speed)

- **Test Case:** Evaluate the usability, form UI, input validation, and layout during data entry.
- **Instructions:** Provide the tester with "Mock Patient Profile A" (Age: 58, BP: 150/95, High Cholesterol: > 240mg/dL, Max Heart Rate: 88 bpm, Chest Pain: Atypical Angina, Fasting Blood Sugar: > 120 mg/dL, Blocked Vessels: 1, Thalassemia: Reversible Defect, No Interventions). Instruct them to enter this data into the system and record the time taken.

Feature 3: Output Interpretation

- **Test Case:** Ensure the Bayesian calculation and AI Explainer are presented clearly.
- **Instructions:** Ask the tester to review the percentage result and the "Aligned AI Explainer." Ask them what the number means to them, how they think the system arrived at the conclusion, and if they desire more explanation.

Feature 4: Using the "What-If" Feature

- **Test Case:** Test the discovery and usability of interactive AI scenarios.
- **Instructions:** Ask the user to determine how starting "Statin therapy" would change the patient's current risk profile using the tool's interactive features.

Feature 5: Understanding the Breakdown Metrics

- **Test Case:** Evaluate the user's comprehension of the mathematical and clinical metrics.
- **Instructions:** Instruct the user to scroll down to the "Risk Factor Breakdown" and "Model Clinical Verification" sections. Ask if it is clear which factors increased or lowered the risk (green/red bars) and if they understand what the verification section is telling them about the AI's accuracy.

Feature 6: Missing Data / Error Handling

- **Test Case:** Test the robustness of the Bayesian network and form logic.
- **Instructions:** Instruct the user to submit the form with the cholesterol data field left completely blank to see if the system updates the probability successfully using partial data or if it crashes.

Features to avoid testing

- **None:** Testers are free and encouraged to explore all available interactive elements of the prototype.

Signoff

I hereby accept this final product. (Yes/No): _____

Signature: _____

Name, Position, Organisation: _____

Date: _____